THOUFIKUR RAHAMAN Y

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Professional Summary

Electrical & Electronics Engineering student with practical exposure to embedded systems, IoT, computer vision and software development through academic laboratories, projects, internships and hackathons. Familiar with Arduino, ESP32, Raspberry Pi, Embedded C, sensor interfacing and serial communication, alongside Python-based application development and database integration. Interested in Embedded Systems, Firmware, IoT and Software Engineering internships.

Education

Sri Venkateswara College of Engineering – B.E. Electrical & Electronics Engineering ***Sept 2024 – Jun 2028***
CGPA: 8.39/10 (Up to 4th Semester)

Internship

Student Intern–Schwing Stetter, Sriperumbudur

- Gained practical exposure to electrical design and electrical control systems in an industrial environment.
- Studied components and applications including relays, contactors, MPCBs, MCBs/MCCBs, 24V DC solenoid valves, proximity sensors and VFDs.

Student Intern–Chennai Metro RailLimited, Chennai

- Gained exposure to metro power distribution, transformers, relay protection, switchgear, circuit breakers and busbar systems.
- Observed overhead traction power systems, pantograph operations, SCADA and electrical safety practices.

Projects

Smart Attendance System – iCUBE Hackathon Winner

- Developed an IoT-enabled, period-based attendance application using camera-based face recognition, supporting timetable-driven automatic mode and manual attendance.
- Implemented face-processing and liveness-validation workflows using OpenCV and MediaPipe; stored attendance records in SQLite and supported reporting.
- **Tools:** Python, OpenCV, MediaPipe, SQLite, Raspberry Pi.

Smart Track Safety System – SIH Hackathon Project

- Designed an ESP32- and sensor-based safety-monitoring system intended to identify abnormal or broken overhead-line conditions.
- Developed the fault-detection and alert mechanism concept, combining electrical safety requirements with embedded monitoring.
- Tools: ESP32, Sensors, Embedded Systems.

Adaptive Quiz AI Platform

- Developed an adaptive testing platform with backend logic that adjusts question difficulty using accuracy and response time.

Diabetic Retinopathy Detection System

- Developed an AI model for retinal fundus-image analysis using deep learning and lesion detection.

Technical Skills

Programming :	C,C++,Embedded C Java,Python(Basic).
Embedded&IoT :	Arduino,ESP32, RaspberryPi, Sensors, Microcontrollers.
Embedded Concepts :	GPIO, ADC, PWM, Timers, Interrupts, Registers, Memory, Bitwise Operations.
Communication :	UART, I2C, SPI, Sensor Interfacing, Serial Communication .
Hardware & Tools :	Multimeter, Circuit Troubleshooting, Arduino IDE, Tinkercad, KiCad, MATLAB.
Industrial:	PLC, SCADA, TIAPortal, Industrial Automation
EngineeringTools :	AutoCAD, MATLAB, Tinkercad, EPLAN, Fusion360, KiCad
Design & Media :	Canva,Adobe Express,Capcut,Social Media Content,Graphic Design
Software / AI :	OpenCV, YOLOv8, EfficientNetB3, ONNX,SQLite, FastAPI, React, Vite, Tailwind CSS

Leadership & Activities

Chief Design Officer, E-Cell – SVCE | Executive Director, Science Club – SVCE | Deputy Secretary, BIS Students Chapter – SVCE